Robótica de Manipulação : 5th Problems Class

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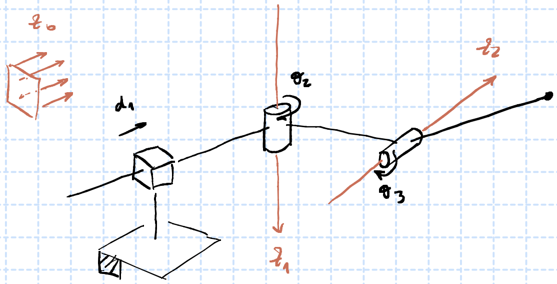
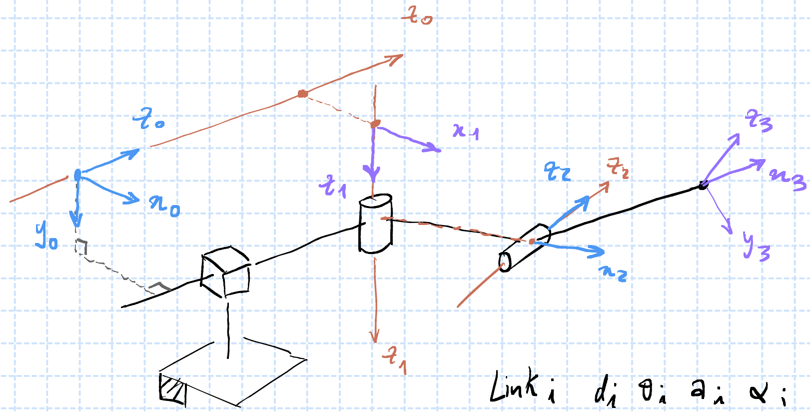
Note Title 2021-04-08

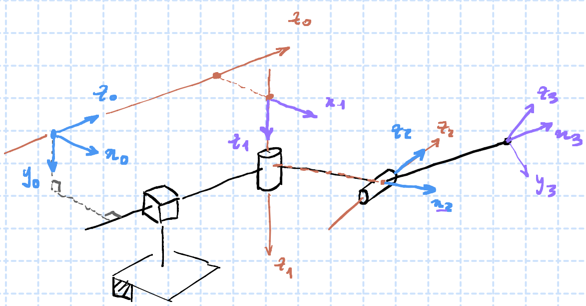
1. (a)



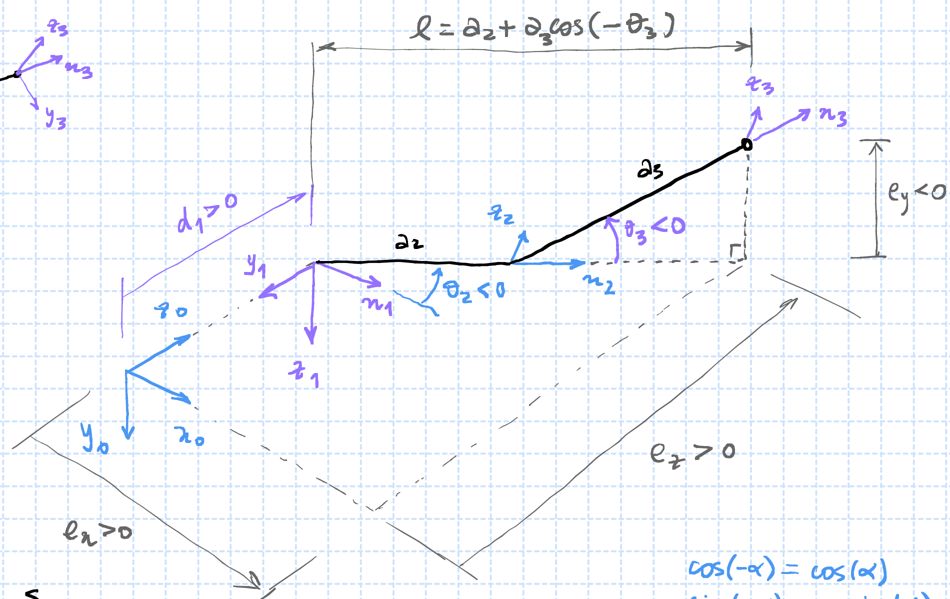


Link i	d_i	θ_i	a_i	α_i
1	d_1	0	a_1	$-\pi/2$
2	d_2	θ_2	a_2	$\pi/2$
3	0	θ_3	a_3	0

(b) Direct Kinematics (Graphical Method)



choose $a_1 = d_2 = 0$



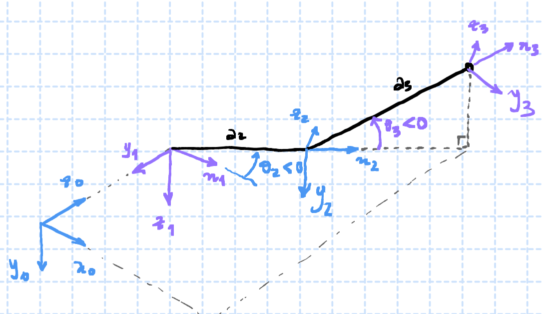
$$P_3 = \begin{pmatrix} e_x \\ e_y \\ e_z \end{pmatrix}$$

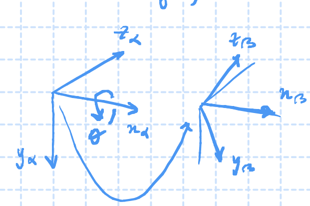
$\cos(-\alpha) = \cos(\alpha)$
$\sin(-\alpha) = -\sin(\alpha)$

$$\begin{cases} -e_y = a_3 \sin(-\theta_3) \Rightarrow e_y = a_3 s_3 \\ e_x = l\cos(-\theta_2) = (a_2 + a_3 c_3) c_2 \\ e_z = d_1 + l\sin(-\theta_2) = d_1 - (a_2 + a_3 c_3) s_2 \end{cases}$$

$$p^\alpha = R^\alpha_\beta \, p^\beta$$

from β to α (direction of coordinates change)





from α to β (direction of motion)

$$R^0_3 = R^0_1 R^1_2 R^2_3 \, ,$$

$$R^0_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} , \quad R^1_2 = \begin{pmatrix} \cos(-\theta_2) & 0 & -\sin(-\theta_2) \\ -\sin(-\theta_2) & 0 & -\cos(-\theta_2) \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} c_2 & 0 & s_2 \\ s_2 & 0 & -c_2 \\ 0 & 1 & 0 \end{pmatrix} ,$$

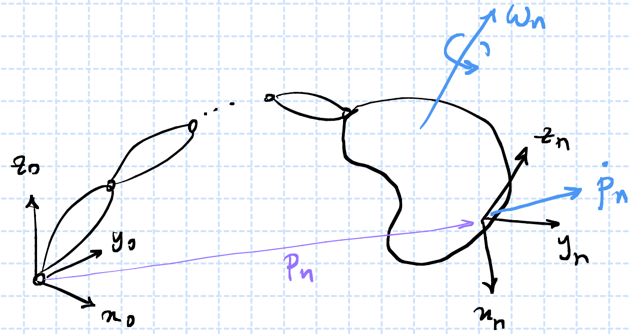
$$R_3^2 = \begin{pmatrix} \cos(-\vartheta_3) & \sin(-\vartheta_3) & 0 \\ -\sin(-\vartheta_3) & \cos(-\vartheta_3) & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} c_3 & -s_3 & 0 \\ s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$R_2^0 = R_1^0 R_2^1 = \begin{pmatrix} c_2 & 0 & s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & c_2 \end{pmatrix}, \quad R_3^0 = R_2^0 R_3^2 = \begin{pmatrix} c_2 c_3 & -c_2 s_3 & s_2 \\ s_3 & c_3 & 0 \\ -s_2 c_3 & s_2 s_3 & c_2 \end{pmatrix}.$$

$$A_3^0 = \left(\begin{array}{c|c} R_3^0 & p_3 \\ \hline 0 & 1 \end{array}\right).$$

(d) Differential Kinematics

$$\boxed{\begin{gathered} \dot{p}_n = \mathcal{J}_p \dot{q}, \quad \omega_n = \mathcal{J}_o \dot{q} \\ \begin{pmatrix} \dot{p}_n \\ \hline \omega_n \end{pmatrix} = \underbrace{\begin{pmatrix} \mathcal{J}_p \\ \hline \mathcal{J}_o \end{pmatrix}}_{\text{geometric Jacobian, } \mathcal{J} \in \mathcal{M}_{6\times n}} \begin{pmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{pmatrix} \end{gathered}}$$



• Position Jacobian, $\quad p_n = p_n(q_1, \dots, q_n), \quad \frac{dp_n}{dt} = \frac{\partial p_n}{\partial q_1}\frac{dq_1}{dt} + \dots + \frac{\partial p_n}{\partial q_n}\frac{dq_n}{dt}$

$$\boxed{\dot{p}_n = \underbrace{\left(\frac{\partial p_n}{\partial q_1} \; \vdots \; \dots \; \vdots \; \frac{\partial p_n}{\partial q_n} \right)}_{\mathcal{J}_p} \begin{pmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{pmatrix}}$$

$$p_3 = \begin{pmatrix} (a_2 + a_3 c_3) c_2 \\ a_3 s_3 \\ d_1 - (a_2 + a_3 c_3) s_2 \end{pmatrix}$$

$$\mathcal{J}_p = \begin{pmatrix} \frac{\partial p_3}{\partial d_1} & \frac{\partial p_3}{\partial \vartheta_2} & \frac{\partial p_3}{\partial \vartheta_3} \end{pmatrix} = \left(\begin{array}{c|c|c} 0 & -(a_2 + a_3 c_3) s_2 & -a_3 c_2 s_3 \\ 0 & 0 & a_3 c_3 \\ 1 & -(a_2 + a_3 c_3) c_2 & a_3 s_2 s_3 \end{array}\right)$$

$$\frac{\partial}{\partial x}\left[\cos(x)\right] = -\sin(x)$$

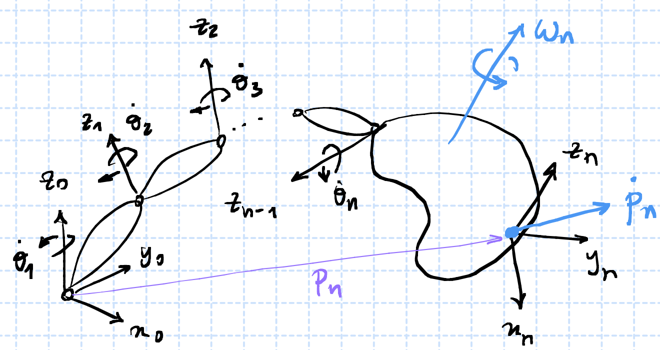
$$\frac{\partial}{\partial x}\left[\sin(x)\right] = \cos(x)$$

• Orientation Jacobian,

$$\omega_n = \omega_{0,n} = \omega_{0,1} + \omega_{1,2} + \dots + \omega_{n-1,n}$$

$$\omega_n^0 = \dot{\vartheta}_1 z_0^0 + \dot{\vartheta}_2 z_1^0 + \dots + \dot{\vartheta}_n z_{n-1}^0$$

$$\omega_n = \underbrace{\begin{pmatrix} z_0^0 & z_1^0 & \cdots & z_{n-1}^0 \end{pmatrix}}_{\mathcal{J}_o} \begin{pmatrix} \dot{\vartheta}_1 \\ \dot{\vartheta}_2 \\ \vdots \\ \dot{\vartheta}_n \end{pmatrix}$$



$$\boxed{\mathcal{J}_o = \begin{pmatrix} \mathcal{J}_{o_1} & \mathcal{J}_{o_2} & \cdots & \mathcal{J}_{o_n} \end{pmatrix}, \quad \mathcal{J}_{o_i} = \begin{cases} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, & \text{if } q_i \text{ is prismatic} \\ z_{i-1}^0, & \text{if } q_i \text{ is revolute} \end{cases}}$$

$$R^0_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} x^0_1 & y^0_1 & \boxed{z^0_1} \end{pmatrix}$$

$$R^0_2 = R^0_1 R^1_2 = \begin{pmatrix} c_2 & 0 & s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & c_2 \end{pmatrix} = \begin{pmatrix} x^0_2 & y^0_2 & \boxed{z^0_2} \end{pmatrix}$$

$$R^0_3 = R^0_2 R^2_3 = \begin{pmatrix} c_2c_3 & -c_2s_3 & s_2 \\ s_3 & c_3 & 0 \\ -s_2c_3 & s_2s_3 & c_2 \end{pmatrix} = \begin{pmatrix} x^0_3 & y^0_3 & z^0_3 \end{pmatrix}$$

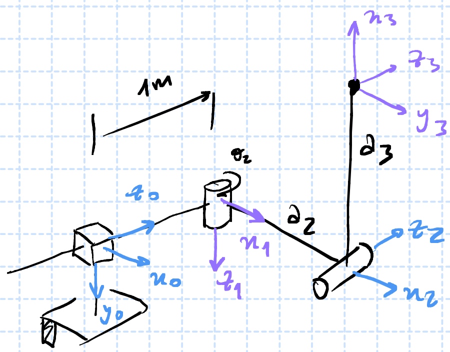
$$\boxed{R^\alpha_\beta = \begin{pmatrix} x^\alpha_\beta & y^\alpha_\beta & z^\alpha_\beta \end{pmatrix}}$$

$$J_o = \left(\begin{array}{c|c|c} 0 & & \\ 0 & z^0_1 & z^0_2 \\ 0 & & \end{array}\right)$$

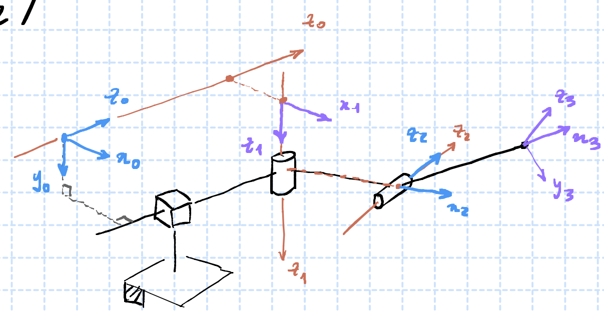
$$J_o = \left(\begin{array}{c|c|c} 0 & 0 & s_2 \\ 0 & 1 & 0 \\ 0 & 0 & c_2 \end{array}\right)$$

$$J = \begin{pmatrix} J_P \\ \hline J_o \end{pmatrix}$$

Project Questions,

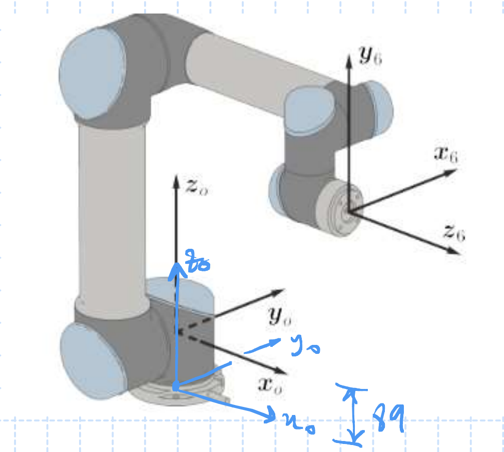


$$q = \begin{pmatrix} 1 \\ 0 \\ -\pi/2 \end{pmatrix}$$



$$p_3 = \begin{pmatrix} a_2 \\ -a_3 \\ d_1 \end{pmatrix}, \quad R^0_3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$0 \approx 1 \times 10^{-15}$$



$$p_6 = \begin{pmatrix} 474 \\ -109 \\ 330 + 89 \end{pmatrix}$$